

09 — Data Persistence Layer & Dataset Ingestion

■ ■ Storage Architecture Overview

The Maritime Port Intelligence Platform utilizes an **In-Memory Analytical Data Pool** powered by **Pandas DataFrames**, backed by persistent CSV files and an HTTP Multipart File Upload service.

While production maritime platforms often connect to PostgreSQL or TimescaleDB, this application uses a high-performance Python time-series ingestion engine that loads, cleans, and indexes multi-year CSV datasets directly into memory upon server initialization.

[illegible]

■ Dataset Schemas & Attributes

1. Primary Dataset: *CODE_PH_0*

- **Source:** Official monthly cargo traffic records mapped to New Mangalore Port Authority (NMPA).
- **Record Count:** 1,354 historical monthly rows.
- **Time Span:** Covers 2021, 2023, 2024, 2025, and 2026.
- **Column Breakdown:**

Column Name	Data Type	Meaning / Purpose	Example Value
<i>CODE_PH_0</i>	String	Name of the calendar month	<i>CODE_PH_0, CODE_PH_1</i>
<i>CODE_PH_0</i>	Integer	Calendar year of record	<i>CODE_PH_0, CODE_PH_1, CODE_PH_2</i>
<i>CODE_PH_0</i>	String	Direction of trade flow	<i>CODE_PH_0</i> (Export) / <i>CODE_PH_1</i> (Import)
<i>CODE_PH_0</i>	String	Commodity category name	<i>CODE_PH_0, CODE_PH_1, CODE_PH_2</i>

<i>CODE_PH_0</i>	Float	Number of ship calls for commodity in current month	<i>CODE_PH_0, CODE_PH_1</i>
<i>CODE_PH_0</i>	Float	Number of ship calls for commodity in same month last year	<i>CODE_PH_0, CODE_PH_1</i>
<i>CODE_PH_0</i>	Float	Primary Metric: Tonnage handled in current month	<i>CODE_PH_0, CODE_PH_1</i>
<i>CODE_PH_0</i>	Float	Tonnage handled in same month of previous year	<i>CODE_PH_0, CODE_PH_1</i>
<i>CODE_PH_0</i>	Float	Year-over-Year percentage volume change	<i>CODE_PH_0, CODE_PH_1</i>

2. Benchmark Dataset: *CODE_PH_0*

- **Source:** Multi-port benchmark dataset used for comparative analytics.
- **Record Count:** 62 weekly records.
- **Ports Covered:** Mumbai, Chennai, JNPT, Kandla, Vizag, Kochi.
- **Column Breakdown:** *CODE_PH_0, CODE_PH_1, CODE_PH_2, CODE_PH_3, CODE_PH_4, CODE_PH_5, CODE_PH_6*.

■ Data Lifecycle: How Data Reaches the Dashboard

1. **Storage:** File remains safely stored on disk in *CODE_PH_0*.
2. **Read & Clean:** *CODE_PH_0* invokes *CODE_PH_1*, strips whitespace from commodity strings, and parses numeric columns.
3. **Resampling:** *CODE_PH_0* creates a continuous month-by-month time series (*CODE_PH_1* to *CODE_PH_2*), filling missing commodity-month pairs with zero volume.
4. **NMPA Infrastructure Mapping:** Enriches each commodity with NMPA facility details (*CODE_PH_0*):
 - Berth assignments (*CODE_PH_0, CODE_PH_1, CODE_PH_2*).
 - Draft depth limits (*CODE_PH_0, CODE_PH_1, CODE_PH_2*).
 - Hinterland consumer linkages (*CODE_PH_0, CODE_PH_1, CODE_PH_2, CODE_PH_3*).
5. **REST API Delivery:** FastAPI serializes data frames into JSON payloads (*CODE_PH_0*).
6. **Frontend Display:** Streamlit renders metrics, Plotly time-series lines, and data quality cards (*CODE_PH_0, CODE_PH_1, CODE_PH_2*).